

FRACTIONAL GAUSSIAN FIELDS ON THE VICSEK FRACTAL

MASTER'S THESIS DEFENCE

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Vicsek Fractal

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$$K = \bigcup_{i=1}^5 \mathcal{S}_i(K), \quad \text{where} \quad \mathcal{S}_i(x) = \frac{1}{3}(x - p_i) + p_i, \quad i = 1, 2, \dots, 5.$$

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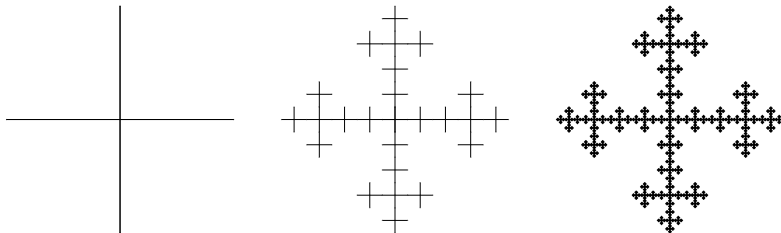


Figure: Vicsek graph, G_0 , G_2 and G_5 .

Hausdorff Dimension

Definition 1.6

Let X be a metric space. If $K \subset X$ and $s \in [0, \infty)$ we define

$$H_\delta^s(K) = \inf \left\{ \sum_{i=1}^{\infty} \text{diam}(U_i)^s : \bigcup_{i=1}^{\infty} U_i \supseteq K, \text{diam}(U_i) < \delta \right\}.$$

The s -dimensional Hausdorff measure is then given by

$$H^s(K) = \lim_{\delta \rightarrow 0} H_\delta^s(K).$$

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Definition 1.8

The Hausdorff dimension, denoted $d_h(K)$ is given by

$$d_h(K) = \inf \{s \geq 0 : H^s(K) = 0\} = \sup \{s \geq 0 : H^s(K) = \infty\}.$$

Hausdorff Dimension

Theorem 1.9

Suppose the open set condition holds for the similarities $\{\mathcal{S}_i\}_{1 \leq i \leq m}$ on $\mathbb{R}^n$ with ratios $0 < c_i < 1$. If, for K the following holds

$$K = \bigcup_{i=1}^m \mathcal{S}_i(K)$$

then $d_h = s$ where s is given by

$$\sum_{i=1}^m c_i^s = 1.$$

Moreover, for this value of s , $0 < H^s(K) < \infty$.

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Open Set Condition: $V \supset \bigcup_{i=1}^m \mathcal{S}_i(V)$, $\mathcal{S}_i(V)$ disjoint

Spectral and Walk Dimension

Walk Dimension: B_t Brownian motion on K

$$c_1 r^{d_w} \leq \mathbb{E} [\tau(x, r)] \leq c_2 r^{d_w},$$

where $\tau(x, r) = \inf \{t \geq 0 : B_t \notin B(x, r)\}$.

Spectral and Walk Dimension

Walk Dimension: B_t Brownian motion on K

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Spectral Dimension: $d_s = \frac{2d_h}{d_w}$

Hausdorff, spectral and Walk Dimension of the Vicsek set

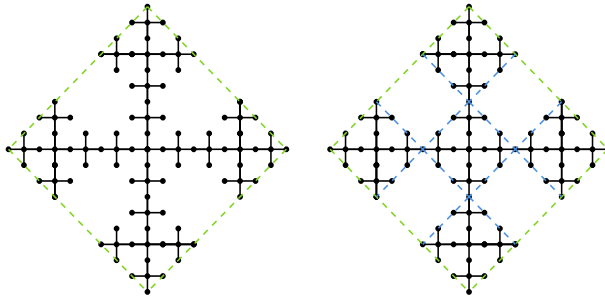


Figure: The opens sets V (green) and $\mathcal{S}_i(V)$ (blue).

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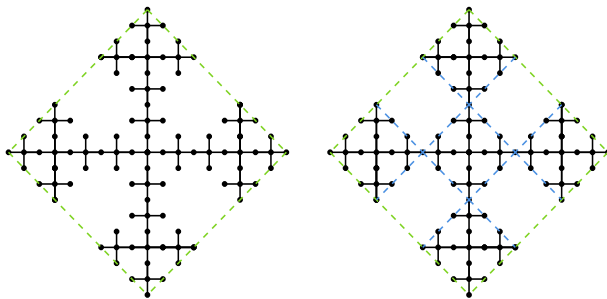


Figure: The opens sets V (green) and $S_i(V)$ (blue).

$$|\mathcal{S}_i(x) - \mathcal{S}_i(y)| = \left| \frac{1}{3}(x - p_i) + p_i - \left(\frac{1}{3}(y - p_i) + p_i \right) \right| = \frac{1}{3}|x - y|.$$

Hausdorff, spectral and Walk Dimension of the Vicsek set

Lemma 1.14

The following constants hold for the Vicsek set K

$$d_h = \frac{\log 5}{\log 3}, \quad d_w = d_h + 1, \quad \text{and} \quad d_s = \frac{d_h}{2d_w} = \frac{\log 5}{2 \log 15}.$$

Normalized Hausdorff Measure

Definition 1.15

The normalized Hausdorff measure on K is given by the Borel measure μ on K , such for any $(i_1, \dots, i_n)$ where $1 \leq i_j \leq 5$, we have

$$\mu(S_{i_1} \circ \dots \circ S_{i_n}(K)) = 5^{-n}.$$

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Definition 1.16

Let $(\mu_m)_{m \geq 0}$ be a sequence of discrete measures on V_m given by

$$\mu_m = \frac{1}{a_m} \sum_{p \in V_m} \delta_p,$$

where $a_m = 5^m \cdot 4 + 1$.

$$\lim_{m \rightarrow \infty} \int_K f \, d\mu_m = \int_K f \, d\mu, \quad \forall f \in C(K)$$

Discrete Laplacian

Definition 2.1

Let $\ell(V_m)$ be the set of functions $f: V_m \rightarrow \mathbb{R}$. Then for any $f \in \ell(V_m)$ the discrete Laplacian on V_m is defined as

$$\Delta_m f(p) = 3^m \frac{1}{N_p} \sum_{q \in V_{m,p}} (f(q) - f(p)), \quad p \in V_m \setminus V_0$$

where

$$N_p = \begin{cases} 5^{-m}, & \text{if } \#V_{m,p} = 4 \\ 2 \cdot 5^{-(m+1)}, & \text{if } \#V_{m,p} = 2 \\ 5^{-(m+1)}, & \text{if } \#V_{m,p} = 1. \end{cases}$$

Kigami Laplacian

Definition 2.2

Let $u \in \text{dom } \mathcal{E}$ and let $f \in C(K)$, then $u \in \text{dom } \Delta_\mu$ and $\Delta_\mu u = f$ if

$$\mathcal{E}(u, v) = \int_K f v \, d\mu, \quad \forall v \in \text{dom}_0 \mathcal{E}.$$

We call Δ_μ the Kigami Laplacian.

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Problem: Hard to find $\Delta_\mu u$

$$\text{dom}_0 \mathcal{E} = \{u \in \ell(V^*) : \mathcal{E}(u) < \infty, \, u = 0, \text{ on } V_0\}$$

$$\mathcal{E}(u) = \lim_{m \rightarrow \infty} r^{-m} \sum_{x \sim y} (u(x) - u(y))^2$$

Kigami Laplacian

Theorem 2.3

Let $u \in \text{dom } \Delta_\mu$. Then the pointwise formula

$$\Delta_\mu u(x) = \lim_{m \rightarrow \infty} r^{-m} \left(\int_K \psi_x^{(m)} d\mu \right)^{-1} \tilde{\Delta}_m u(x) \quad (1)$$

holds with the limit uniform across $V_* \setminus V_0$. Conversely, suppose u is a continuous function and the right side of (1) converges uniformly to a continuous function on $V_* \setminus V_0$. Then $u \in \text{dom } \Delta_\mu$ and (1) holds.

Dirichlet Laplacian

- Recall:

$$\text{dom}_0 \mathcal{E} = \{u \in \ell(V^*) : \mathcal{E}(u) < \infty, u = 0, \text{ on } V_0\}$$

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- Exists operator Δ :

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- Δ a unique densely defined self-adjoint operator.
 - extension of the Kigami Laplacian Δ_μ
 - The operator Δ with domain $\text{dom } \Delta$ is the Dirichlet Laplacian.

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- Eigenvalues $0 < \lambda_1^m \leq \lambda_2^m \leq \dots \leq \lambda_{N_m}^m$ of $-\Delta_m$

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- $(\phi_i^m)_{1 \leq i \leq N_m}$ be the corresponding eigenfunctions.

Definition 2.4

Let $s \geq 0$, we then define the discrete Riesz kernel on V_m as

$$G_s^m(x, y) = \sum_{i=1}^{N_m} (\lambda_i^m)^{-s} \phi_i^m(x) \phi_i^m(y), \quad \text{for } x, y \in V_m.$$

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Definition 2.5

For $f \in \ell(V_m)$ and $s \geq 0$, the discrete fractional Laplacian $(-\Delta_m)^s: L^2(V_m, \mu_m) \rightarrow L^2(V_m, \mu_m)$ is defined by

$$(-\Delta_m)^{-s} f(x) = \frac{1}{a_m} \sum_{y \in V_m} G_s^m(x, y) f(y)$$

Semigroups and Heat Kernels

Theorem 2.6

If A is a densely defined self adjoint operator on $\mathcal{H}$ then it is the generator of the strongly continuous self-adjoint contraction semigroup on $\mathcal{H}$ defined as $P_t = e^{-tA}$.

Semigroups and Heat Kernels

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If A is a densely defined self adjoint operator on $\mathcal{H}$ then it is the generator of the strongly continuous self-adjoint contraction semigroup on $\mathcal{H}$ defined as $P_t = e^{-tA}$.

Definition 2.7

Let $(P_t)_{t \geq 0}$ be a strongly continuous self-adjoint contraction semigroup on $L^2(X, \mu)$. We say that the semigroup $(P_t)_{t \geq 0}$ admits a heat kernel if there exists a measurable function

$p: \mathbb{R}_{>0} \times X \times X \rightarrow \mathbb{R}_{\geq 0}$ such that for every $t > 0$, a.e. we have

$$P_t f(x) = \int_K p_t(x, y) f(y) d\mu(y)$$

with $x, y \in X, f \in L^\infty(X, \mu)$.

Preliminary to Fractional Laplacian

- Recall: Δ with $\text{dom } \Delta$ is densely defined self-adjoint operator

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- P_t admits heatkernel $p_t(x, y)$

Theorem 2.8

There exists an orthonormal basis $\{\phi_i\}_{i=1}^{\infty}$ of $L^2(K, \mu)$ with ϕ_j being eigenfunctions of $-\Delta$. If λ_i is the eigenvalue belonging to ϕ_i , then $\lambda_i > 0$ and $\lim_{i \rightarrow \infty} \lambda_i = \infty$. Moreover

$$p_t(x, y) = \sum_{j=1}^{\infty} e^{-\lambda_j t} \phi_j(x) \phi_j(y),$$

where the convergence is uniform for all $t \in (0, \infty)$, $x, y \in K \times K$.

Fractional Laplacian

Definition 2.9

Let $s \geq 0$, and $f \in L^2(K, \mu)$. The fractional Laplacian $(-\Delta)^{-s}$ is defined as

$$(-\Delta)^{-s}f = \sum_{j=1}^{\infty} \frac{1}{\lambda_j^s} \phi_j \int_K \phi_j(y) f(y) d\mu(y).$$

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Definition 2.10

For parameter $s \geq 0$ we define the fractional Riesz kernel G_s by

$$G_s(x, y) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} p_t(x, y) dt, \quad x, y \in K, x \neq y.$$

Space of test functions

Definition 2.11

We define the following space of test functions $S(K)$ as

$$S(K) = \left\{ f \in C_0(K) : \forall k \geq 0 \lim_{n \rightarrow \infty} n^k \left| \int_K \phi_n(y) f(y) d\mu(y) \right| = 0 \right\}$$

with the topology defined by the family of norms

$$\|f\|_k = \left\| (-\Delta)^{-k} f \right\|_{L^2(K, \mu)}.$$

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$$\|f\|_k = \left\| (-\Delta)^{-k} f \right\|_{L^2(K, \mu)}.$$

- $S(K)$ is a nuclear space
- Why do we want it to be a nuclear space?

Important results

Lemma 2.12

Let $s > 0$. For $f \in S(K)$ and $x \in K$

$$(-\Delta)^{-s} f = \int_K G_s(x, y) f(y) d\mu(y).$$

Important results

Lemma 2.12

Let $s > 0$. For $f \in S(K)$ and $x \in K$

$$(-\Delta)^{-s}f = \int_K G_s(x, y) f(y) \, d\mu(y).$$

Corollary 2.13

(i) Let $s > 0$, for $f, g \in S(K)$

$$\int_K (-\Delta)^{-s}f (-\Delta)^{-s}g \, d\mu = \int_K \int_K f(x)g(y) G_{2s}(x, y) \, d\mu(x) \, d\mu(y).$$

(ii) Let $s > 0$ the for $f \in S(K)$

$$(-\Delta)^{-s}f = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} P_t f(x) \, dt.$$

Discrete Fractional Gaussian Fields

Definition 2.14

Let $s \geq 0$. A discrete fractional Gaussian field X_s^m with parameter s on V_m is a Gaussian vector indexed by V_m with mean 0 and covariance matrix $\Sigma = (G_{2s}^m(x, y))_{x, y \in V_m}$.

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Notation: For any $f \in \ell(V_m)$ we will use the notation

$$X_s^m(f) = \frac{1}{a_m} \sum_{p \in V_m} f(p) X_s^m(p).$$

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Proposition 2.15

For $f, g \in \ell(V_m)$ we have that

$$\mathbb{E} [X_s^m(f) X_s^m(g)] = \frac{1}{a_m} \sum_{p \in V_m} (-\Delta_m)^{-s} f(p) (-\Delta_m)^{-s} g(p).$$

Example of Fractional Gaussian Field

Example 2.16

Let $(W_i)_{1 \leq i \leq N_m}$ be white noise. Then define

$$X_s^m(x) = \sum_{i=1}^{N_m} (\lambda_i^m)^{-s} \phi_i^m(x) W_i, \quad x \in V_m.$$

Fractional Gaussian Fields

Definition 2.17

Let $s \geq 0$. A fractional Gaussian field X_s with parameter s on K is a centered Gaussian field, $\{X_s(f) : f \in S(K)\}$, such that for any $f, g \in S(K)$

$$\mathbb{E} [X_s(f)X_s(g)] = \int_K (-\Delta)^{-s} f (-\Delta)^{-s} g \, d\mu.$$

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Theorem 1

Let $s \geq 0$. There exists a centered Gaussian distribution X_s on $S'(K)$ such that for $f, g \in S(K)$

$$\mathbb{E} [X_s(f)X_s(g)] = \int_K (-\Delta)^{-s} f (-\Delta)^{-s} g \, d\mu.$$

Fractional Gaussian Fields Existence

Theorem 2 (Bochner-Minlos)

Assume $\mathcal{X}$ is a nuclear space. Any continuous positive-definite linear functional Λ on $\mathcal{X}$ satisfying $\Lambda(0) = 1$ is the Fourier transform of a countably additive positive normalized measure on $\mathcal{X}'$

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Proof of Existence: The space $S(K)$ is a nuclear space, why it is enough to use Bochner-Minlos on the functional

$$\Lambda: f \rightarrow \exp\left(-\frac{1}{2}\|f\|_k^2\right).$$

Since it gives us a unique probability measure ν such that

$$\exp\left(-\frac{1}{2}\|f\|_k^2\right) = \int_{S'(K)} e^{i\langle x, f \rangle} d\nu(x).$$



fractional Brownian field

Definition 2.18

Let $s > \frac{d_h}{2d_w}$. The fractional Brownian field with parameter s is defined as

$$\tilde{X}_s(x) = \int_K G_s(x, y) dW(y).$$

where W is white noise.

Remark 2.19

Since $s > \frac{d_h}{2d_w}$ then we can write

$$\tilde{X}_s(x) = \sum_{i=1}^{\infty} \lambda_i^{-s} \phi_i(x) W_i,$$

where $W_i = W(\phi_i)$ is white noise.

fractional Brownian field

Proposition 2.20

Let $s > \frac{d_h}{2d_w}$. Then the Gaussian random field defined by

$$X_s(f) = \int_K f(x) \tilde{X}_s(x) d\mu(x), \quad f \in S(K)$$

has the law of a FGF with parameter s . Moreover, if there exists a Gaussian field $(\tilde{Y}_s(x))_{x \in K}$ on K with

$$\mathbb{E} \left[\int_K \tilde{Y}_s(x)^2 d\mu(x) \right] < \infty$$

such that the Gaussian field defined by

$$Y_s(f) = \int_K f(x) \tilde{Y}_s(x) d\mu(x), \quad f \in S(K)$$

has the law of FGF with parameter s , then $s > \frac{d_h}{2d_w}$.

Main Theorem

Lemma 3

For all $f \in S(K)$ and $s \geq 0$ the following holds,

$$\frac{1}{a_m} \sum_{p \in V_m} f_m(p) (-\Delta_m)^{-2s} f_m(p) \rightarrow \int_K f(x) (-\Delta)^{-2s} f(x) d\mu(x)$$

as $m \rightarrow \infty$.

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Main Theorem 2.21

Let $s \geq 0$, then X_s^m converges to X_s in distribution in $S'(K)$ when $m \rightarrow \infty$.

Proof of Main Theorem:

- Since $S(K)$ is a nuclear space.

For all $f \in S(K)$,

$$\mathbb{E} \left[\exp \left(it X_s^m(f_m) \right) \right] \rightarrow \mathbb{E} \left[\exp \left(it X_s(f) \right) \right], \quad \text{as } m \rightarrow \infty$$

where $f_m = f|_{V_m}$.

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where $f_m = f|_{V_m}$.

- Next:

$$\begin{aligned} \mathbb{E} \left[\exp (itX_s^m(f_m)) \right] &= \exp \left(-\frac{t^2}{2} \mathbb{E} \left[(X_s^m(f_m))^2 \right] \right) \\ &= \exp \left(-\frac{t^2}{2a_m} \sum_{p \in V_m} f_m(p) (-\Delta_m)^{-2s} f_m(p) \right). \end{aligned}$$

- Lastly:

$$\begin{aligned}\mathbb{E} [\exp(itX_s(f))] &= \exp \left(-\frac{t^2}{2} \mathbb{E} [(X_s(f))^2] \right) \\ &= \exp \left(-\frac{t^2}{2} \int_K f(-\Delta)^{-2s} f \, d\mu \right)\end{aligned}$$

The proof now follows from 3.

Matrix Representation

Recall, the *discrete Laplacian* on V_m is defined as

$$\Delta_m f(p) = 3^m \frac{1}{N_p} \sum_{q \in V_{m,p}} (f(q) - f(p)), \quad p \in V_m \setminus V_0$$

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The *Laplacian matrix* L_m is given by

$$(L_m)_{i,j} = \begin{cases} \#V_{m,p_i} \cdot 3^m \cdot \frac{1}{N_{p_i}}, & \text{if } i = j \\ -1 \cdot 3^m \cdot \frac{1}{N_{p_i}}, & \text{if } i \neq j \text{ and } p_j \in V_{m,p_i} \\ 0, & \text{otherwise} \end{cases}$$

which satisfies:

$$L_m F = \begin{pmatrix} \Delta_m f(x_1) \\ \vdots \\ \Delta_m f(x_{N_m}) \end{pmatrix}, \quad \text{where } F = \begin{pmatrix} f(x_1) \\ \vdots \\ f(x_{N_m}) \end{pmatrix}.$$

Goal: Show

$$X_s^m(f_m) \rightarrow \tilde{X}_s(f)$$

where $f_m = f|_{V_m}$ and $m \rightarrow \infty$ where

$$X_s^m(x) = \sum_{i=1}^{N_m} (\lambda_i^m)^{-s} \phi_i^m(x) W_i, \quad \text{and} \quad \tilde{X}_s(x) = \sum_{i=1}^{\infty} \lambda_i^{-s} \phi_i(x) W_i,$$

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Problem: Can not simulate $m \geq 7$.

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$$\#V_m = 5^m \cdot 4 + 1, \quad V_8 = 1.562.501$$

